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Exploring Intellectual Property in the Digital Realm: A Bibliometric Study on Research on the Management and Protection of Data-Based Intellectual Property

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Abstract: The management and protection of data-based intellectual property have attracted increasing attention in the academic community due to the rapid development of digital technologies and data-driven industries. However, a comprehensive and multidimensional examination of the research landscape is still required to better understand its structure and evolution. Using CiteSpace software, this study conducts a bibliometric analysis, revealing key trends and patterns in collaboration, co-citation, and keyword co-occurrence in the field of data-based intellectual property. Our findings show a growing body of literature on data IP management, with a significant increase in publications since 2013. We identify that collaboration between regions, especially the United States, China, and the United Kingdom, leads global efforts, but institutional collaboration remains underdeveloped. In terms of co-citation, seminal works by Jaffe, Hall, and Samuelson form the foundation of the current research, while emerging research focuses on technological innovations like blockchain and AI. The analysis further reveals that future research is likely to explore the intersections of data privacy, innovation, and legal frameworks. Compared with previous studies, this paper builds a knowledge framework for data-based intellectual property management from a holistic perspective of bibliometrics, analyses the current challenges, and outlines future research directions, which is of significant reference value to both scholars and practitioners.

Keywords: data; intellectual property; bibliometric; data sharing; privacy



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1. Introduction

The advent of the big data era poses new challenges for intellectual property protection. There is a growing need for innovative data-based IP management strategies in fields such as education, healthcare, and business. The increase in software, digital media, and databases requires sophisticated protections to ensure the fair use of data and clear attribution of IP rights. Advances in technologies such as blockchain and artificial intelligence have reshaped the IP framework and legal updates are required to address the new complexities.

In the field of data management law and policy, a growing body of research explores the intersection between legal frameworks, data-based intellectual property rights, and privacy rights. Harrison examines the differences between data ownership in Europe and the United States, focusing on the legal definition of de facto data ownership [1]. Richards explores the tension between data privacy and freedom of speech as established under the US Constitution, while Boddy proposes an approach to employee data management that balances privacy and intellectual property rights [2,3]. Fisher and Fortmann emphasize the importance of the scientific community's management of the data commons, drawing parallels with the management of natural resources [4]. Kemp discusses the legal infrastructure and argues in favor of a coherent governance strategy [5]. Finally, Leistner and

Antoine highlight the growing data economy in the EU, calling for reforms to improve the portability and accessibility of data [6]. Overall, these studies emphasize the need to reform the legal framework to balance data-based intellectual property rights and privacy rights in the digital age.

In terms of data protection technology, the research has focused on safeguarding user data in an increasingly interconnected digital ecosystem. Zhang JX highlight vulnerabilities in enterprise platforms, addressing risks related to intellectual property infringement in big data environments [7]. Sengupta and Bhadauria propose a watermarking method to provide intellectual property protection for control data flow graphs during behavioral synthesis [8]. Veale studied the threat posed by machine learning model inversion and membership inference attacks under GDPR, highlighting the need for strong data protection strategies [9]. These studies contribute to the development of diverse technological solutions aimed at enhancing data-based intellectual property protection across various sectors.

In specific subject areas, the research has highlighted the role of legal transparency and data protection in personalized medicine, education, and biological research. Mohamad suggests that synthetic electronic health records can both address privacy concerns in research and comply with legal standards [10]. Rolnick advocates for the development of policies that balance data privacy with the societal value of the data, particularly in the context of health data [11]. Similarly, Popescu explores the need to harmonize bespoke regulations regarding personal data protection and intellectual property rights in medical research, in line with the GDPR [12]. Komljenovic criticizes the marketization of education and highlights the need for public intervention to protect intellectual property rights in educational resources [13]. Rajam and van Veen emphasize the importance of protecting the intellectual property rights of data in the development of data-driven educational resources using machine learning [14].

Finally, the relationship between data-based intellectual property rights and the economy is an emerging area of research. Tao and Bhuiyan examine the economic impact of securing big data, weighing the costs of cybersecurity against the benefits of preventing data breaches [15]. Shelby explores the economic and policy challenges in making federally funded research data available to the public while protecting national security and intellectual property rights [16]. Hirth advocates for open data in the energy industry to drive innovation and discusses the legal and economic implications of providing open data on European power systems [17]. Together, these studies emphasize the importance of balancing accessibility, privacy, security, and economic efficiency when using data.

In this study, we use CiteSpace to conduct a comprehensive bibliometric analysis of the management and protection of data-based intellectual property [18]. Specifically, (1) through a collaboration analysis, we investigate the global collaboration networks among authors, institutions, and regions, focusing on their interconnections and overall structures; (2) using co-citation analysis, we identify the most influential authors, journals, and literature in the field, detailing their research focuses and categorizations to guide scholars toward the most critical resources; (3) using a co-occurrence analysis, we explore current research hotspots and anticipate future developments by examining co-occurring keywords and their citation bursts, providing an insight into the field's knowledge trajectory; and (4) based on the above analyses, we establish a knowledge framework that offers a clear and multidimensional overview of the management and protection of data-based intellectual property while also discussing the current challenges and potential future research directions in the field.

2. Materials and Methods

2.1. Research Method: Bibliometric Analysis

Bibliometrics integrates mathematics, statistics, and bibliographies to quantitatively analyze and categorize the extensive corpus of the scholarly literature, providing insights into research patterns and trends. This methodology, established by Pritchard in 1969, has evolved to form the core of information science's logical structure, facilitating quantitative

and intuitive literature analyses through knowledge mapping and cluster analysis [19]. It offers a multidimensional perspective of the development of specific research fields, enabling the identification of research status, trends, and major contributing forces.

In this study, CiteSpace 6.2.R4 was used for the research process analysis, research evolution analysis, and research hotspot analysis [18]. CiteSpace's ability to visualize researcher collaborations, co-citations, and co-occurrences, as well as its ability to identify emergent terminology, provides a dynamic view of the research trends and highlights evolving and emerging frontiers in research on the management and protection of digital intellectual property. VOSviewer (1.6.20) is another software used for bibliometric analysis, which has a unique advantage in the presentation of graphs. This study utilized VOSviewer to present the author collaboration network, because this allowed for a clearly presented graph.

2.2. Data Retrieval

Our data sources were the Web of Science (WoS) and Web of Science core collection. The WoS data are of high quality and authority: the literature indexed in WoS's SCIE and SSCI indices has a high level of scholarly excellence and each indexed article undergoes a rigorous peer review process. This ensured the reliability of our sources and provided a solid foundation for our analyses.

Our study spanned the period from 1996 to 4 May 2024, the date of retrieval. This period was chosen because the first relevant publication on the management and protection of digital IP appeared in 1996.

With a search date of 4 May 2024, the search query (TS = "intellectual propert*" AND TI = "data") AND ((DT = ("ARTICLE" OR "EDITORIAL MATERIAL" OR "PROCEEDINGS PAPER" OR "REVIEW")) NOT (DT = ("EARLY ACCESS") OR DT = ("RETRACTED PUBLICATION")))) retrieved 507 articles from the WoS. Subsequent manual screening and deduplication refined this number to 314 articles, creating a corpus of the relevant literature that accurately reflects the state of research on digital IP from 1996 to 2024 (the file of 314 articles is provided in Supplementary Materials download_314.txt). This dataset formed the basis of our comprehensive bibliometric analysis, which aimed to reveal trends, key contributors, and emerging themes in the field.

2.3. Data Filtering

1. Due to the high level of noise associated with the keyword 'data', manual screening was necessary even after limiting the search to the title field. Three researchers were involved in this phase of the work, focusing on retaining content related to intellectual property (IP) management and the protection of research data.
2. The research team used Citespace to de-duplicate the literature, removing redundant entries and ensuring that the dataset was more streamlined and the analyses more focused.

2.4. Research Procedures

The main processes are specified as follows (Figure 1):

1. Descriptive statistical analysis: Firstly, annual growth trends, key journals, and research categories are analyzed. This method has been used to track the development of data IP research since 1996.
2. Collaboration analysis: Collaboration networks of authors, institutions, and countries are visualized. These networks show how stakeholders collaborate, highlighting the most influential contributors.
3. Co-citation analysis: Node types, such as references, cited authors, and cited journals, are selected to map co-citation patterns. This reveals the foundational literature and key theories that have influenced the field.
4. Keyword co-occurrence analysis: The research's evolution is mapped, and hotspots and turning points are detected through an analysis of keyword co-occurrence networks. This helps to understand emerging trends and shifts in the research.

5. The findings from each analysis are combined to form a comprehensive overview. This integrates the research results into a knowledge framework, predicting future directions and areas of focus in data-based intellectual property research.

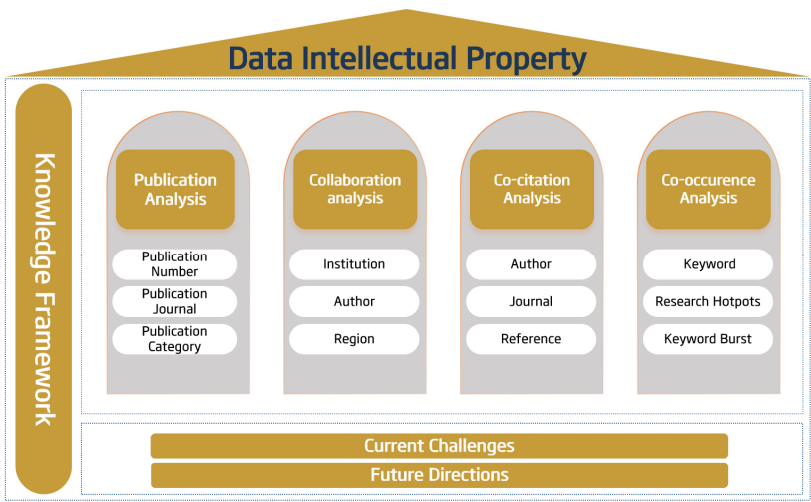


Figure 1. Research process.

3. Results

3.1. Publication Statistics

The aim of the publication analysis was to reveal the dominant publication landscape and its evolution to help researchers gauge the popularity and emergence of the field and to identify the topics and journals that are of greater scholarly interest [20].

3.1.1. Annual Publication Number Analysis

Figure 2 shows that the amount of peer-reviewed publications on data-based intellectual property (IP) grew slowly from 1996 to 2010, with fewer than 10 publications per year. The peaks in 2007 and 2011 show the growing interest in data protection. From 2011 to 2018, the number of publications increased steadily, reaching 17 publications in 2016. In 2019, data-based IP research surged, culminating in 49 publications in 2022. This rapid expansion can be attributed to the increasing significance of data privacy, driven by global regulations such as the General Data Protection Regulation (GDPR), and concerns over data governance and digital monopolies [21]. Many countries introduced or reinforced data protection laws in 2022, encouraging scholars to focus on managing and safeguarding data-based IP [22,23]. Additionally, frequent large-scale data breaches and cyberattacks in 2022 heightened public and business concerns about data security, spurring academic interest in exploring effective intellectual property management to protect data privacy. At the same time, the emergence of COVID-19 may have been responsible for stimulating research into the development of the digital economy and data-based intellectual property rights. However, after peaking in 2022, the number of publications declined to 23 in 2023 and 15 in 2024, likely due to shifts in research priorities and the resolution of some pressing issues. The lower number of publications in 2024 is due to the period of data collection only extending until May.

3.1.2. Publication Number Analysis

Table 1 presents the top ten journals with the most publications on the management and protection of data-based intellectual property, indicating the most active journals in this field. Several key observations can be drawn from the data: (1) The journal *Lecture Notes in Computer Science* leads in terms of publication volume, with 16 papers. The publications in *Lecture Notes in Computer Science* and *Multimedia Tools and Applications* span from 1997 to 2005. While numerous, the articles are relatively outdated, reflecting

an earlier period of research. *Computer Law & Security Review* and *IEEE Transactions on Information Forensics and Security* have recent publications on these topics and are actively publishing; these journals had the most publications after 2011. *Electronic Library* and *Program-Electronic Library and Information Systems* were active from 2004 to 2011, but have declined since then. The *Journal of Documentation* and *Learned Publishing* showed a steady rate of publication over a longer period, from 2000 to 2020 (refer to Supplementary Files: *Co-citation Journals Publications*). These journals cover a broad range of topics within computer science, making them a fitting platform for research on data-based intellectual property, especially in areas intersecting with technology. Similarly, *Multimedia Tools and Applications*, which ranks second, with 11 papers, caters to the technological aspects of data handling and intellectual property. These journals' broad scopes allow for diverse discussions, ranging from theoretical frameworks to practical applications in data-based IP management. (2) There is notable variation in the focus of the top journals. For instance, *Computer Law & Security Review* and *Electronic Library* emphasize the importance of legal and informational perspectives in this field. These journals focus on the intersection of law, data security, and library science, highlighting the interdisciplinary nature of research on data-based intellectual property.

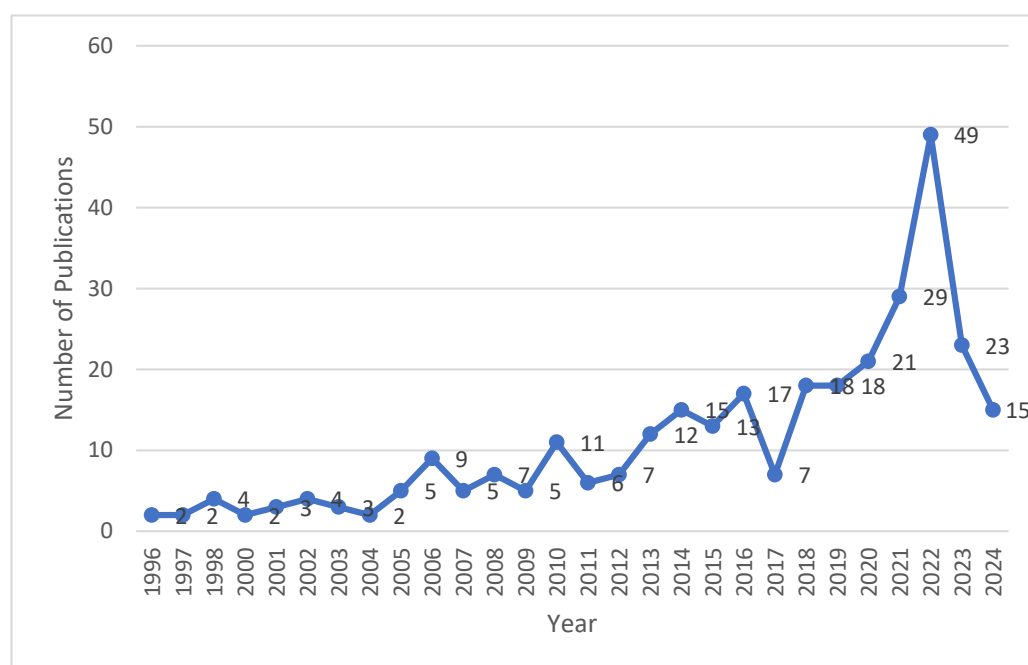


Figure 2. Annual publication statistics.

Table 1. Publication journal statistics.

Ranking	Journal	5-Year IF	Count
1	Lecture Notes in Computer Science	0.302	16
2	Multimedia Tools and Applications	2.9	11
3	Computer Law Security Review	3.4	7
4	Electronic Library	1.8	7
5	IEEE Transactions on Information Forensics and Security	7.3	7
6	IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems	2.9	6
7	Program Electronic Library and Information Systems	1.102	6
8	Journal of Documentation	1.8	5
9	Learned Publishing	2.6	5
10	Online Information Review	3.3	5

In conclusion, this analysis helps researchers identify influential journals for future submissions and encourages them to target high-impact publications; for example, publication in *IEEE Transactions on Information Forensics and Security* is recommended for technical contributions, while *Computer Law & Security Review* should be considered for legal perspectives on data IP management.

3.1.3. Publication Category Analysis

Table 2 lists the top ten categories in data-based intellectual property management research. These top ten categories span across technology (computer science, information systems, engineering, electrical and electronic), law (law), business (management, business), and interdisciplinary fields (multidisciplinary sciences, information science, and library science). Additionally, some more detailed findings were obtained. (1) Importance and influence are not necessarily the same. Specifically, the category “management” has the highest centrality (0.66), followed by “computer science, information systems” (0.36), “economics” (0.17), and “engineering, electrical and electronic” (0.19), which suggests that the management, technology, and economic categories have significant influence and are dominant in this field. Centrality measures the importance of a category within a network, indicating its ability to connect different nodes and serve as a bridge between research areas. A high centrality value signifies that the category exerts a substantial influence and plays a pivotal role in shaping the research landscape. This highlights the significant influence and dominance of the management, technology, and economics categories in this field. However, the category “computer science, information systems” has the highest co-occurrence counts (48) but its centrality is lower than that of “management”, indicating that while information systems are central to data-based IP management research, their influence on management practices is greater than their influence on the purely technological aspects. (2) The earliest category to be found was “law” (1997), demonstrating that scholars have been concerned about legal frameworks in data-based IP management for a long time. For instance, representative studies have explored the impact of legal policies on data protection and the enforcement of intellectual property rights in the digital age. Additionally, the research highlights that the integration of legal considerations is crucial for developing comprehensive data-based IP management strategies that align with technological advancements and economic incentives [24,25]. (3) The technology-related categories (computer science, engineering) are a focal point of the research. Among these categories, “computer science, information systems” appears relatively early (1998) and holds a high centrality (0.36), following the management and law categories, indicating that scholars have continuously emphasized the technological aspects of data IP management. For instance, recent studies explore the role of advanced technologies, such as artificial intelligence and big data analytics, in protecting intellectual property and managing data rights in complex digital environments; other studies further investigate the intersection of technology and law, focusing on how technological innovations necessitate evolving legal frameworks to effectively safeguard intellectual property [25].

Table 2. Publication category analysis.

Ranking	Count	Centrality	Year	WoS Categories
1	48	0.36	1998	Computer Science, Information Systems
2	34	0.03	1997	Law
3	32	0.19	1998	Engineering, Electrical & Electronic
4	32	0.17	1997	Economics
5	29	0.66	2001	Management
6	22	0	2002	Computer Science, Theory & Methods
7	19	0	2001	Computer Science, Software Engineering
8	17	0	2002	Multidisciplinary Sciences
9	16	0	2006	Business
10	13	0.15	2005	Information Science & Library Science

3.2. Cooperation Relations

Collaboration analysis offers a clear representation of collaborative networks, revealing the relationships among researchers to provide an overview of the structure of the network and key research areas. This approach is also useful in identifying major contributors, such as institutions, regions, and authors, to better understand the current state and development of collaborations [26]. By highlighting influential collaborators, it encourages more involvement from less active scholars and points to potential partnerships in the field [20,27]. The following analysis will present both broad research trends and specific details, enabling readers to obtain a comprehensive understanding of the field's characteristics and providing insight into more niche areas of this research.

3.2.1. Institutional Collaboration Analysis

Institutional cooperation analysis helps us to learn the relevance of the partnerships among institutions in research on the management and protection of data-based intellectual property. Table 3 shows the top ten institutions with the most frequent collaborations. These institutions are spread across three continents: North America (4), Europe (3), and Asia (3). This shows that data-based intellectual property management research collaborations are widespread. Specific details are as follows: The University of California is the leading collaborator. Specifically, it has the highest collaboration count (11). Although it has the most collaborations, its low centrality indicates that its leadership position in the network is not prominent. The University of California's research direction is focused on the governance of data commons, ethical considerations of data privacy and sovereignty, and the advancement of scientific research across various fields, such as medicine, personal health information, geographic information, and scientific data management [4,28,29]. (2) The University of London has eight collaborations and the same centrality of 0.01, suggesting a similar pattern of active participation but limited influence in the overall collaboration network. The latest representative research from this university focuses on the development of a homomorphic encryption method for genotypes and phenotypes, enabling secure collaborative genomic analyses while maintaining data confidentiality [30]. The institutions in this field have not yet established close and effective collaborative relationships.

Table 3. Institutional collaboration analysis.

Ranking	Institutions	Region	Count	Centrality	Year
1	University of California System	United States	11	0.01	1997
2	University of London	United Kingdom	8	0.01	2013
3	National Institute of Technology (NIT System)	India	5	0	2022
4	University of California Los Angeles	United States	5	0	2003
5	Harvard University	United States	5	0.01	2002
6	Max Planck Society	Germany	4	0	2008
7	University of Illinois System	United States	4	0	2006
8	Chinese Academy of Sciences	China	3	0	2014
9	Chinese University of Hong Kong	China	3	0	2022
10	Consejo Superior de Investigaciones Científicas (CSIC)	Spain	3	0	2014

In sum, these findings provide us with some ideas when it comes to identifying potential institutions to collaborate with. Researchers can search for potential collaborators that match their research interests, as different institutions have different research focuses.

The institution collaboration network clusters are shown in Figure 3; these clusters contain 345 nodes and 382 links. The cluster analysis in CiteSpace is typically employed to group research topics or concepts with similar characteristics, allowing for the differentiation of these groups with distinct labels, thereby highlighting the structure of a specific research domain. Our specific findings are as follows: (1) Research on “#0 mobile health” in the field of data-based intellectual property primarily focuses on data privacy, security,

intellectual property rights, data-sharing, compliance with regulations, and ethical considerations. The leading collaboration institutions include the University of California, Los Angeles (UCLA), and the University of California System. The most recent representative research investigates the multidimensional legal aspects of personal genomic sequence data in South African law, focusing on property rights, personality rights, and intellectual property rights, and examines their potential for ownership and the need for harmonization [31]. (2) Some institutions focus on cluster #1, non-communicable diseases, such as the notable collaborations between the University of London and Cornell University. This cluster involves a significant amount of data handling, where protecting intellectual property rights regarding the patient data and research outcomes is crucial. In this type of research, institutions are likely to focus on balancing the need for data-sharing in research with the necessity of protecting patient privacy and data ownership rights. The collaboration outlines a set of guidelines developed to address the complex demands on healthcare institutions when sharing clinical data [32]. They also investigate how competing narratives of data ownership in health research influence data-sharing practices, emphasizing that ethical and social concerns often outweigh formal legal ownership when shaping biomedical data governance [33]. The “#2 profitability” cluster, led by Harvard University and Johnson and Johnson, examines the economic aspects of data-based IP, particularly how protected health data can contribute to profitability in the healthcare sector. (3) The cluster “#3 energizing economy” is directly relevant to the topic of the management and protection of data-based intellectual property. The leading collaboration institutions in this cluster include the University of Colorado Boulder and the University of Colorado System. These institutions primarily collaborate on research on the economic implications of data-driven innovations, particularly in the energy and environmental sectors. Protecting intellectual property in this context is vital for ensuring that data assets are secure and can be effectively leveraged to drive economic growth. (4) The cluster “#4 vegetation dynamics”, led by institutions such as Aarhus University and the Senckenberg Gesellschaft für Naturforschung, focuses on environmental and ecological research. In the context of data-based intellectual property, this cluster explores the management and protection of large datasets related to vegetation and environmental changes. Ensuring the protection of intellectual property in this domain is crucial as ecological data are becoming increasingly important for climate change research and conservation efforts. The most recent representative study proposes a model for managing intellectual property rights in the West African Vegetation Database that incentivizes data-sharing while allowing the owners of the data to retain control over access [34]. (5) The collaboration cluster “#5 biotechnology industry” is also prominent, as represented by the research of Bayer AG and Erasmus University Rotterdam. This cluster is focused on the biotechnology sector, where managing and protecting intellectual property is vital to the development and commercialization of new therapies and technologies. Collaboration in this cluster focuses on developing secure data-sharing frameworks that balance the need for collaboration with the safeguarding of proprietary information in preclinical safety and pharmacology studies.

3.2.2. Author Collaboration Analysis

Table 4 outlines the top 10 collaborative authors in the field of data-based intellectual property management. Generally, the frequency of collaboration among authors is modest, with all listed authors contributing to two collaborative publications each, and all having a centrality value of 0. This suggests a low level of interconnectedness within the network, indicating that collaborations tend to occur in isolated groups rather than across a broader, more integrated network.

Figure 4 shows some collaboration groups, although external collaboration among these groups is relatively weak. Bidmeshki, Jin, and Dutta form a tight collaboration network, indicating frequent cooperation on related research topics. The team focus on advancing intellectual property protection and data secrecy in technology development. They introduced and enhanced the proof-carrying hardware intellectual property (PCHIP)

framework, focusing on automating security proof processes, preventing malicious data modifications, and ensuring trustworthiness in hardware designs through mechanisms like VeriCoq-IFT and VeriCoq-H, emphasizing the protection of sensitive information and the integrity of third-party IPs [35,36].

Another notable team is formed by Jana, Biswapati, and Chakraborty, who also form a strong collaborative network. This group might focus on data protection or data-sharing frameworks, as their collaboration suggests an ongoing exchange of ideas and methodologies that could contribute to the development of more robust and secure data management practices. This team focuses on developing advanced data hiding and image-processing techniques to enhance multimedia security [37–39].

The cooperation group headed by Briggs, Katharine focuses on improving drug development through enhanced data-sharing in the pharmaceutical industry, addressing legal and intellectual property challenges [40].

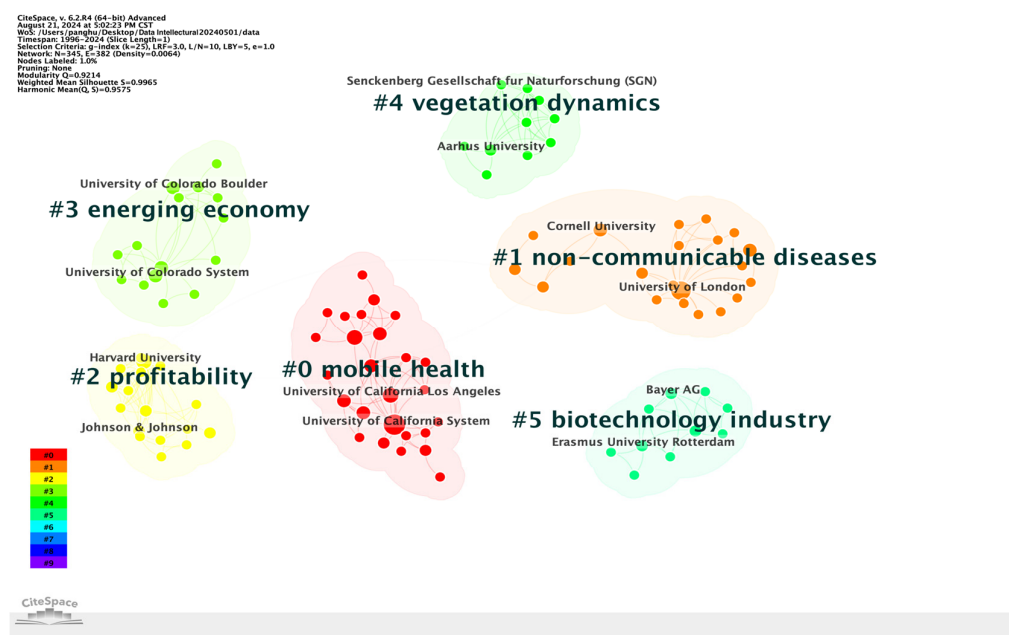


Figure 3. Institutional collaboration networks.



Figure 4. Author collaboration network.

Table 4. Top 10 collaborative authors.

Ranking	Institutions	Count	Centrality	Year
1	Guo, Xiaolong	2	0	2017
2	Li, Teng	2	0	2022
3	Dutta, Raj Gautam	2	0	2017
4	Bidmeshki, Mohammad-Mahdi	2	0	2017
5	Jin, Yier	2	0	2017
6	Assuncao, Larissa Santos	2	0	2022
7	Jana, Biswapati	2	0	2024
8	Wang, Sha	2	0	2022
9	Briggs, Katharine	2	0	2014
10	Zhang, Yu	2	0	2022

3.2.3. Regions Collaboration Analysis

Table 5 presents the top ten regions that are actively collaborating in the field of data-based intellectual property and protection. The data reveal that collaboration is concentrated in both developed and emerging economies, with significant contributions from regions such as the USA, China, and Europe. Here are some detailed observations: The United States leads collaboration in this field, ranking first, with 89 collaborations and a centrality of 0.37, indicating its strong influence and central role in global collaborations. The USA's leadership in this area began in 1997, showcasing its long-standing engagement in data-based IP and protection issues. The latest collaborative effort aims to improve genomic research collaboration. Homomorphic encryption is used to protect genetic and phenotypic data. This keeps the data private and follows the FAIR principles in agricultural genome-to-phenome research [30]. China ranks second, with 72 collaborations and a centrality of 0.22, highlighting its notable influence in the field of data-based intellectual property since 2003. Despite having fewer collaborations than the USA, China's significant centrality reflects its growing role in global intellectual property discussions. The latest collaboration focuses on the management and protection of intellectual property in biomedical data, highlighting the impact of the re-use of scientific datasets on research outcomes and the benefits of data-sharing in enhancing academic reputation and citations [41]. England ranks third but boasts the highest centrality (0.57), indicating its significant impact on collaborative networks despite its having fewer total collaborations (47) than the USA and China. England's contributions, starting from 2005, focused on creating comprehensive legal frameworks for data protection and intellectual property, particularly in the context of the European Union's stringent regulations. The latest collaborative efforts identify unresolved legal issues and the underexplored aspects of data protection in medical 3D printing, highlighting the need for clearer legislation and terminological consistency to address intellectual property and liability concerns [42]. India is fifth, with 17 collaborations and a centrality of 0.2; its significant contributions began in 2007. India's involvement reflects its growing tech industry and the need to protect the intellectual property generated by its rapidly expanding digital economy. Research from India often centers on the need to balance IP protection with the accessibility of technology and fostering innovation in developing markets. The most recent research developed a blockchain-based intrusion detection system for IoT city data, which incorporates device authentication and collaborative detection to protect against DDoS attacks [43].

Figure 5 illustrates the regional collaborative networks, containing 57 nodes and 182 connections. The clusters reveal distinct primary research interests across various regions, highlighting the focus and intensity of collaboration in specific areas.

England and Spain tend to collaborate on “#0 big data”. Collaborative efforts in this cluster are focused on developing legal frameworks, data governance policies, and technologies to protect and utilize large datasets effectively [44]. The collaboration between the Netherlands and Canada focuses on “#1 privacy”, particularly the protection of personal data within the context of data-based IP [45–47]. Research in this cluster addresses the

challenges of maintaining data privacy in a world that is increasingly reliant on digital technologies, with an emphasis on developing privacy-preserving techniques and legal protections that align with international standards. The USA and China collaborate on “#2 profitability”, emphasizing the need to balance profitability and public benefit when managing big data as a public resource. Their collaboration proposes a framework that allows for controlled access to pooled personal data for public purposes, while protecting privacy, and suggests that companies are incentivized to share data voluntarily through regulatory sandboxes and safe harbors [48]. Cluster #3, involving Belgium and Germany, centers on the broader aspects of intellectual property rights. Germany’s involvement emphasizes its role in driving policy and legal reforms in IP protection. The collaboration between India and Saudi Arabia is concentrated on “#4 data hiding techniques”, which are crucial for protecting sensitive information and maintaining data security. This cluster investigates advanced steganography methods, encryption techniques, and the application of these methods to protect intellectual property and ensure the confidentiality of data in various digital applications. The latest research presents a robust data hiding scheme using Block Truncation Coding (BTC) and Hamming code, which is designed to securely conceal data within highly compressed images, enhancing protection against various attacks and improving intellectual property rights and digital security [39].

Table 5. Top 10 collaborative regions.

Ranking	Region	Count	Centrality	Year
1	USA	89	0.37	1997
2	PEOPLES R CHINA	72	0.22	2003
3	ENGLAND	47	0.57	2005
4	GERMANY	26	0.23	2006
5	INDIA	17	0.2	2007
6	NETHERLANDS	16	0.14	2006
7	CANADA	15	0.04	2006
8	AUSTRALIA	12	0.03	2008
9	SPAIN	11	0.01	2006
10	ITALY	10	0.02	2008

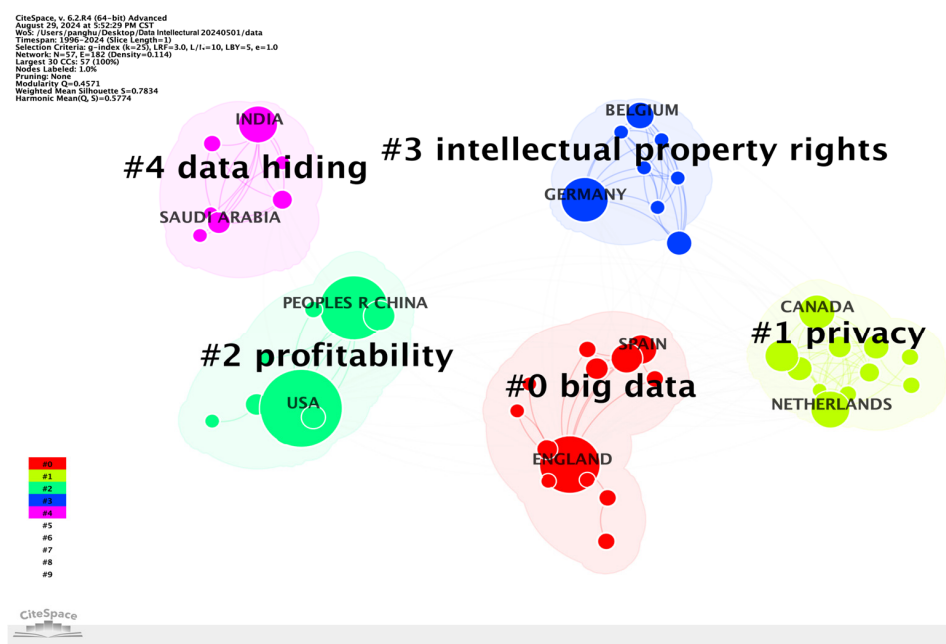


Figure 5. Regional collaboration network.

3.3. Co-Citation Status

The co-citation analysis in CiteSpace serves to uncover the relationships and connections among authors, journals, and publications within a research domain. Specifically, when two documents, authors, or journals are frequently cited together by other publications, it suggests a meaningful correlation, implying that they are contributing to the same research area or academic topic. The frequency of these co-citations reflects the strength of their association, indicating a shared impact or focus within the academic community. By conducting co-citation analysis, researchers can identify key trends, influential works, and foundational studies that shape the current state of the field. Importantly, the co-cited references highlight pivotal research that may not be directly included in the dataset but still plays a critical role in guiding ongoing scholarship and identifying the core literature that has significantly influenced the research landscape.

3.3.1. Author Co-Citation Analysis

Table 6 highlights the top 10 most frequently co-cited authors in the field of data-based intellectual property's management and protection. These authors represent central figures within this academic field. The co-citation analysis revealed key authors whose works are frequently referenced together, indicating their significance and influence within the research community. In terms of co-citation, all authors weighted the same. Here are the detailed findings: Jaffe AB is the most frequently co-cited author, with 16 co-citations, followed by Hall BH, with 11, and Samuelson P, with 9. These authors represent central figures within this academic field. Jaffe AB's research evolved over time: in 1993, he studied how knowledge spreads locally, finding that domestic patents often cite other domestic patents [49]. By 1999, he had shifted to international knowledge flows, looking at how patent citations differ across countries and the role of national borders [50]. In 2000, he analyzed changes in the U.S. patent system over the past 20 years and their impact on innovation [49,51]. Schwartz PM has the highest centrality, at 0.19, suggesting that this author plays a more significant role in connecting various subfields or bridging knowledge areas, despite having only eight co-citations. His work is influential in the academic and policy spheres, as he has also advised international organizations including the European Commission and contributed to the development of data privacy principles. His books *Information Privacy Law* and *Privacy Law Fundamentals* are particularly noteworthy. Acemoglu D also has a high centrality (0.18), showing their influence within the network with only seven co-citations. His latest work proposes a mediated data-sharing scheme to address the issues of low data prices and excessive sharing [52]. In addition, we can observe that the centrality scores of the other co-cited authors are comparatively modest and uniform (all below 0.1), suggesting that their influence in this area is comparatively limited. Overall, this analysis helps us to identify potential key influencers that we should watch out for in this field in the future.

Figure 6 shows the author co-citation network, containing 975 nodes and 3277 connections. We clustered these nodes so that multiple nodes correspond to a representative theme. The figure shows nine main clusters that demonstrate the primary research focuses.

The co-citation clusters within the realm of data-based intellectual property and data ownership reveal a rich tapestry of specialized areas, each contributing to the broader understanding of how data rights are managed and protected in the digital age.

In the interdisciplinary fields, “#0 Biotechnology Industry” and “#1 University Patents” highlight the significance of IP in fostering innovation. The former examines the impact of IP laws on biotech innovation and commercialization, with key authors like Calabresi G and McManis C. The latter, featuring the authors Jaffe AB, Hall BH, and Aghion P, discusses the translation of academic research into patented technologies and the economic impact of university patents.

The legal, philosophical, and strategic aspects are encapsulated in “#3 Ownership” “#5 Non-Practicing Entities”, and “#7 Scientific Integrity”. Cluster #3, featuring Lemley MA, Samuelson P, and Schwartz PM, discusses the evolving definitions of data ownership

and the intersection with privacy and IP law. Cluster #5, with Arora A, Smith PJ, and Bessen J, scrutinizes the role of NPEs in the IP landscape and their impact on innovation and litigation. Cluster #7, led by Benkler Y, Burk Danl, and Reichman JH, addresses the integrity of scientific research and the protection of IP rights, emphasizing the balance between open science and proprietary rights.

Table 6. Top 10 most co-cited authors.

Ranking	Count	Centrality	Year	Cited Authors
1	16	0.05	1997	JAFFE AB
2	11	0.03	1997	HALL BH
3	9	0.03	1997	SAMUELSON P
4	8	0.02	2019	LEMLEY MA
5	8	0.19	2016	SCHWARTZ PM
6	8	0.08	2005	AGHION P
7	7	0.01	1997	GRILICHES Z
8	7	0.06	2009	MOWERY DC
9	7	0.18	2012	ACEMOGLU D
10	6	0.01	2021	EUROPEAN COMMISSION

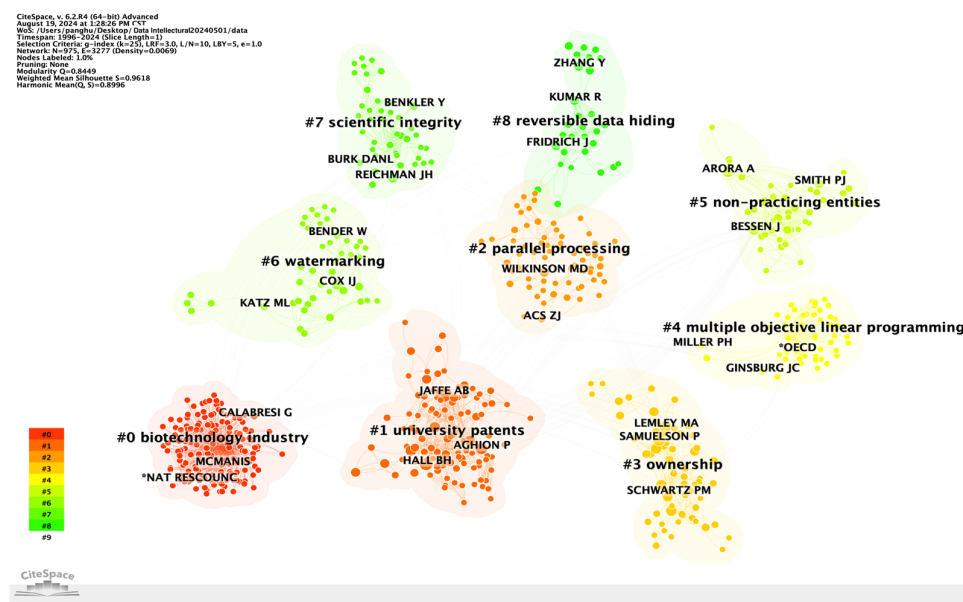


Figure 6. Author co-citation network. (An asterisk (*) before an author's name indicates that the author is an organization, not a person).

“Cluster #8 Reversible Data Hiding” introduces methods that allow for the embedding of information in a manner that supports the full restoration of the original data post-extraction; this field is led by scholars such as Zhang Y, Kumar R, and Fridrich J. “Cluster #6 Watermarking”, featuring core authors like Katz ML, Cox LJ, and Bender W, investigates the use of digital watermarks to protect content and prevent unauthorized use. “Cluster #2 Parallel Processing” explores the patenting and legal considerations of parallel computation innovations; research in this area is spearheaded by Wilkinson MD and ACS ZJ. “Cluster #4 Multiple Objective Linear Programming” applies mathematical optimization to IP management, with contributions from Miller PH, Ginsburg JC, and OECD, focusing on strategic IP portfolio management and licensing.

3.3.2. Journal Co-Citation Network

The “most co-cited journals” are the journals that appear together in the references of multiple papers in our dataset. Looking at this can help us to identify the most influential

and foundational journals within our field. The journal co-citation analysis focusing on the management and protection of data-based intellectual property revealed several influential journals. Table 7 presents the top ten co-cited journals that significantly contribute to research in this domain. These journals are frequently referenced together in the academic literature, indicating their relevance and impact. Some detailed findings are as follows: (1) *Research Policy* ranks first, with 61 co-citations and a centrality of 0.06, highlighting its importance as a foundational resource in intellectual property research. It has a high number of publications but a relatively low centrality, suggesting that its influence is concentrated within specific research clusters. Although it is frequently cited, its role as a connector in the citation network is limited. This suggests that the journal primarily serves a specific field or audience, making a significant contribution within its niche but having less impact as a bridge between broader areas of research. (2) *American Economic Review (AM ECON REV)* is notable for its high centrality score of 0.37, despite its having fewer citations (45) than *Research Policy*. This suggests that while it may not be the most cited, it plays a crucial role in connecting various research streams, making it a pivotal publication in understanding economic perspectives of intellectual property. (3) *Science*, with 42 co-citations and a centrality of 0.11, has a strong influence, particularly in the scientific and technical dimensions of data protection and intellectual property. (4) *PLOS ONE* and *IEEE Access*, both relatively recent entries from 2018 and 2019, respectively, have quickly become significant, particularly *IEEE Access*, with 25 citations, and *PLOS ONE*, with 26 citations, reflecting their growing importance in discussions on the intersection of technology, data, and intellectual property. (5) *Nature* maintains its prominence, with 24 citations and a centrality of 0.12, showing its ongoing relevance in broad scientific research, including research on intellectual property management and data rights.

Table 7. Top 10 most co-cited journals.

Ranking	Count	Centrality	Year	Cited Journals
1	61	0.06	1997	RES POLICY
2	45	0.37	1997	AM ECON REV
3	42	0.11	2003	SCIENCE
4	26	0.05	2018	PLOS ONE
5	25	0.03	2019	IEEE ACCESS
6	24	0.12	2000	NATURE
7	23	0.11	1998	LECT NOTES COMPUT SC
8	22	0.03	2019	TECHNOL FORECAST SOC
9	21	0.05	1996	INTELLECTUAL PROPERT
10	21	0.02	2001	REV ECON STAT

From 1996 to 2024, the co-citation trends among the top ten journals have shown distinct patterns, with the notable increase in recent years being attributed to the proliferation of Open Access (OA) platforms. During the initial period from 1996 to 2005, co-citation activity was relatively low for most journals, with only occasional spikes, such as that in *Nature*, which peaked at 3 in 2005, and *Review of Economic Statistics*, which peaked at 2 in the same year. At this stage, OA journals had not yet made a significant impact. The period between 2006 and 2015 saw a steady increase in co-citation activity for traditional journals. In particular, *Science* and *Research Policy* showed a steady increase, with *Science* reaching a peak of 4 in 2014 and *Research Policy* peaking at 5 in both 2008 and 2014. The *Review of Economic Statistics* also peaked at 4 in 2014. This period marked a transitional phase in which the co-citation counts of established journals began to stabilize. From 2016, the impact of Open Access journals became more prominent. The OA journal *PLOS ONE* continued to be cited at a high rate, peaking at 5 between 2021 and 2023. Technology-focused OA platforms such as *Lecture Notes in Computer Science* and *IEEE Access* also reached their co-citation peaks, with *Lecture Notes in Computer Science* peaking at 9 in 2020 and *IEEE Access* peaking at 5 in 2021. *Technological Forecasting and Social Change*, another OA journal, peaked at 6 in 2021. These trends show the increasing importance of OA journals in fostering

academic discourse, particularly in the fields of technology and interdisciplinary research. In summary, the data indicate a transition from sporadic early peaks to a sustained increase in co-citation activity. OA journals have increasingly reached peak co-citation figures in recent years (refer to Supplementary Files: Annual Publications).

The co-citation journal clusters portrayed in the analysis illustrate the primary research themes related to the management and protection of data-based intellectual property (Figure 7). The network, which contains 959 nodes and 3827 connections, reveals eight significant clusters: “#0 topic modeling”, “#1 common pool resources”, “#2 data hiding”, “#3 business models”, “#4 smartphone”, “#5 psychiatry”, “#6 indigenous data sovereignty”, “#7 controlled vocabularies”, and “#8 scientific and technical data and information”. These clusters represent the key areas of scholarly focus in this field.

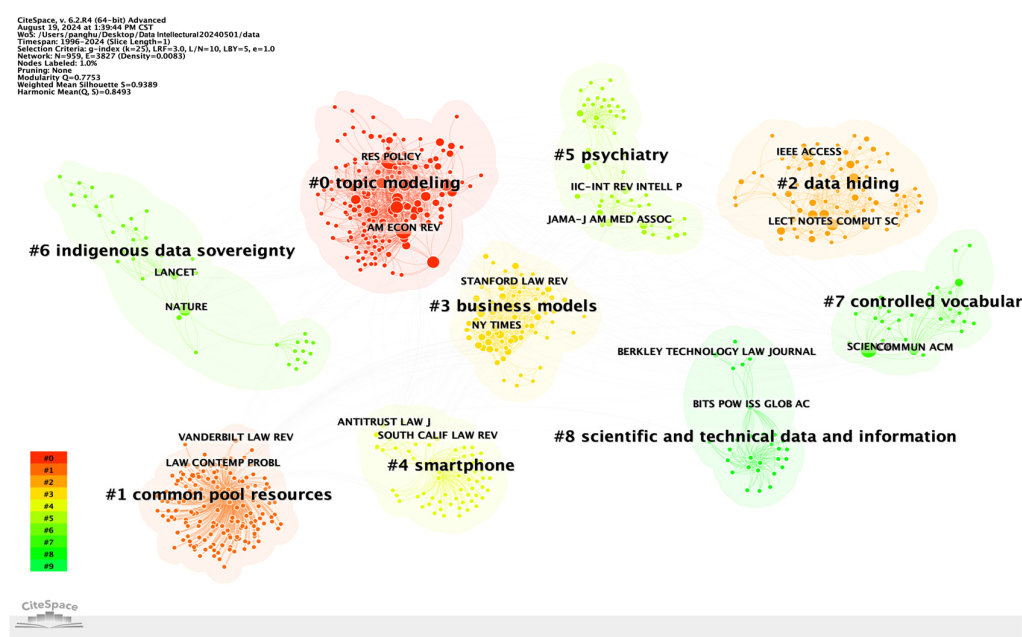


Figure 7. Journal co-citation network.

Research on the management and protection of data-based intellectual property (IP) can be broadly categorized into several directions. “#0 topic modeling” and “#3” focus on the intersection of IP with economics and business strategy”. Journals such as *Research Policy* and *American Economic Review* emphasize how data-driven insights can inform IP policy and drive innovation, while journals such as *Stanford Law Review* explore the strategic management of IP in the digital economy. Aspects of strategic management such as “#1 common pool resources” and “#4 smartphone” focus on the legal aspects of data-based IP, with journals such as the *Vanderbilt Law Review* and the *Antitrust Law Journal* examining governance, open access, and conflicts in laws related to shared data resources, particularly in the areas of smartphone technology and antitrust issues. “#2 data hiding” and “#8 scientific and technical data and information” focus on technological solutions to privacy issues, with journals such as *IEEE Access* and *Global Access* discussing the use of technologies such as encryption and digital watermarking to protect intellectual property and promote innovation. “#5 psychiatry” and “#6 indigenous data sovereignty” address the ethical and cultural dimensions of data-based intellectual property, with *JAMA* and *The Lancet* highlighting the challenges of managing sensitive medical data and local data. Finally, cluster “#7 controlled vocabularies” looks at the role of controlled vocabularies in scientific data management, with journals such as *Science Communication* emphasizing the importance of using standard terminology to improve data-sharing and protection.

3.3.3. Reference Co-Citation Network

Table 8 highlights the top 10 most frequently co-cited references. The analysis of reference co-citations is used to identify seminal works that have significantly influenced the field.

One of the key clusters in this area is the application of information and communication technology (ICT) to agricultural data management. Wolfert S makes a significant contribution to this area, discussing how ICT can improve the management of agricultural data and the protection of related intellectual property rights [53]. Similarly, the FAIR (Findable, Accessible, Interoperable, and Reusable) data principles emphasized by Wilkinson MD are crucial [54]. Regarding data protection, Adi Y proposed a method of watermarking through the backtracking of deep neural networks to protect data [55]. Digital multimedia copyright protection is another critical area of data-based intellectual property rights management, with T. D. Sairam proposing an improved high-capacity data hiding scheme that utilizes pixel value adjustment and modulus operation to conceal data [56]. In the context of international trade, Ivus O concluded that stronger patent rights increase high-tech exports to the developing world [57]. In the area of cultural economics, Giles P points out that companies translate human experience into data, which are then used to generate profits for these companies. Data are collected not only to “know our behavior” but also to “shape our behavior” [58].

Table 8. Top 10 most co-cited references.

Ranking	Count	Centrality	Year	Cited References
1	4	0.06	2017	Wolfert S, 2017, AGR SYST, V153, P69, DOI 10.1016/j.agsy.2017.01.023 [53]
2	4	0.37	2016	Wilkinson MD, 2016, SCI DATA, V3, P0, DOI 10.1038/sdata.2016.18 [54]
3	3	0.11	2004	Mowery DC, 2004, IVORY TOWER AND INDUSTRIAL INNOVATION [59]
4	3	0.05	2018	Adi Y, 2018, PROCEEDINGS OF THE 27TH USENIX SECURITY SYMPOSIUM, V0, P1615 [60]
5	3	0.03	2002	Campbell EG, 2002, JAMA-J AM MED ASSOC, V287, P473, DOI 10.1001/jama.287.4.473 [55]
6	3	0.12	2019	Giles P, 2019, J CULT ECON-UK, V12, P612, DOI 10.1080/17530350.2019.1639068 [58]
7	2	0.11	2020	Sairam TD, 2020, MULTIMED TOOLS APPL, V79, P17003, DOI 10.1007/s11042-019-7557-9 [56]
8	2	0.03	2010	Ivus O, 2010, J INT ECON, V81, P38, DOI 10.1016/j.jinteco.2009.12.002 [57]
9	2	0.05	2004	Scotchmer S, 2004, INNOVATION AND INCENTIVES, V0, P0 (https://mitpress.mit.edu/9780262693431/innovation-and-incentives/ (accessed on 10 October 2024))
10	2	0.02	2003	Agrawal R, 2003, VLDB J, V12, P157, DOI 10.1007/s00778-003-0097-x [61]

Figure 8 portrays the co-citation clustering of the critical literature in the field of data-based intellectual property management, including 3105 nodes and 9058 connections. The co-cited references cover various significant themes in this domain. (1) The most prominent cluster is “intellectual property rights” (#0), where foundational works by Reichman JH (1997, 1993) laid the groundwork for understanding how intellectual property law governs the management and protection of data. This body of research was central to subsequent studies that address the balance between data ownership and accessibility [62]. Reichman introduced the concept of compensatory liability rules, balancing the relationship between intellectual property protection and the public interest to protect first-time creators from mass-copying. Nachbar points out that intellectual property restrictions do not apply to other powers in Article I of the Constitution, and argues that the exclusive power of intellectual property is just another form of regulation used by Congress to place economic rents on particular interests. (2) Another notable cluster, “reifying data” (#1), includes works by Nachbar T (2004) and Reichman JH (2003), which explore the transformation of data into valuable assets through legal frameworks [63–65]. This cluster plays a crucial role in

guiding how data are treated as property within different legal systems. (3) In “data privacy reconciliation” (#2), significant works by Balkin JM (2004) and Volokh E (2005) address the tensions between data privacy and the commercial use of personal information [66,67]. Balkin JM discusses the protection of free speech in the digital age. Volokh E discusses the concept of whether states should have constitutional sovereign immunity in copyright and patent lawsuits. These studies are highly relevant as the discussion around privacy protection and data ownership evolves alongside technological advancements. (4) “Data integration” (#4), represented by Campbell EG (2002) and Reichman JH (1999), examines the challenges of integrating various data sources while safeguarding intellectual property, a key theme in research related to big data and cross-border data exchanges [55,62,68]. Campbell EG believes that a lack of resources and scientific priorities can play an important role in scientists’ decisions to withhold data, materials, and information from other academic geneticists. (5) Other notable clusters include “patent rights” (#6) and “geospatial data” (#5), which discuss intellectual property protection in the context of specific data types. “Patent rights”, for instance, examines legal mechanisms to protect innovations, while “geospatial data” highlights the unique challenges in managing spatial information under intellectual property laws. In conclusion, these clusters provide a detailed map of the research landscape in data-based intellectual property management, with foundational references guiding the evolution of knowledge in this field. Scholars can leverage these key references to advance the discourse in emerging areas, such as data privacy, ownership, and integration.

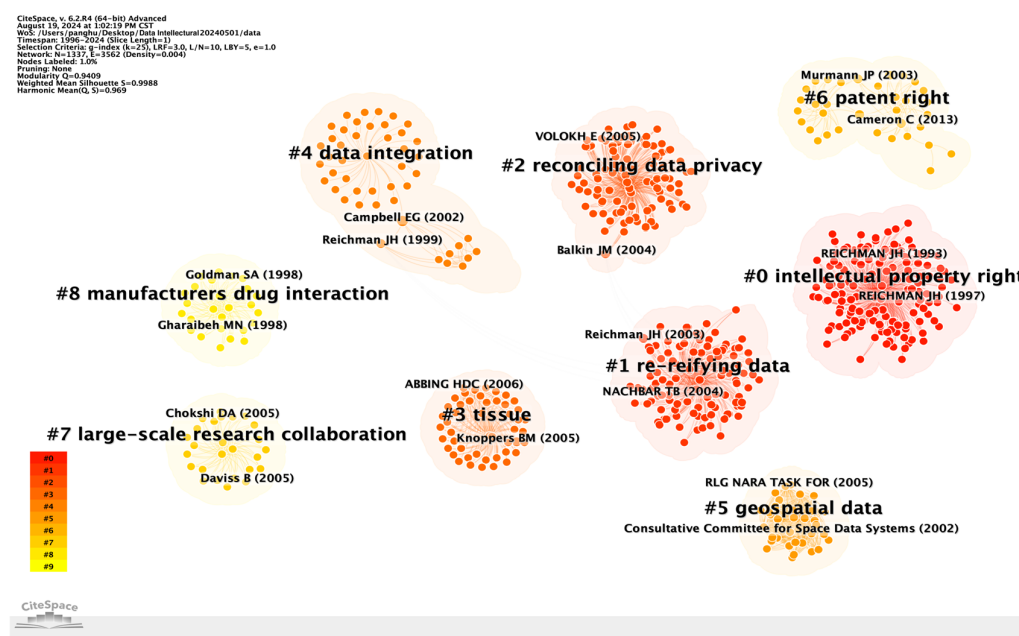


Figure 8. Reference co-citation network [53,54,59].

3.4. Co-Occurrence Analysis

3.4.1. Keyword Co-Occurrence Analysis

Keyword co-occurrence analysis is essential to understand the main topics and evolution of Intellectual Property Rights (IPR) research focusing on data. Table 9 lists the most frequent keywords, revealing the core areas of interest in the field [69].

By analyzing the co-occurrence of keywords in the field of data-based intellectual property, we can identify key themes and emerging research trends. Among them, ‘intellectual property’ has the highest frequency and centrality (85, 0.79), suggesting that it plays a fundamental role in connecting different research themes related to the legal framework, ownership, and enforcement of intellectual property rights in the data context. For example, Mitra-Kahn analyzed how the Australian legal framework governs and protects intellectual

property data, highlighting the impact of intellectual property law on innovation and business ownership and Thaldar explored the legal challenges of genomic data ownership, focusing on the intersection and enforcement of intellectual property rights, privacy, and ownership [31,70].

Table 9. Top 10 co-occurring keywords.

Ranking	Count	Centrality	Year	Keywords
1	85	0.79	2001	intellectual property rights
2	22	0.1	2005	innovation
3	19	0.1	2014	big data
4	12	0.07	2007	technology
5	12	0.07	2016	privacy
6	10	0.04	2001	information
7	10	0.13	2010	data sharing
8	9	0.08	2001	citations
9	8	0.05	1997	law
10	7	0.02	2014	data mining

The keyword ‘innovation’ is in second place, with a volume of 22 and a centrality of 0.1. This link between innovation and IPR is critical, as it highlights the ongoing interest in ensuring that legal protections foster technological progress [71,72]. The centralities of the co-occurring keywords ‘data sharing’ and ‘big data’ are above 0.1, showing that they are prominent keywords that represent the intersection of technological advances and intellectual property rights, particularly in the management and protection of the vast amounts of data generated in the current digital economy. Nicol, D.’s research highlights the need for legal reforms in Europe’s digital policy framework to address data-sharing and the challenges arising from the lack of standardized contracts and the need to protect database creators’ sui generis rights and trade secrets [73]. Similarly, Andanda, P. argues for a shift from ownership to custodianship in the governance of big data in health research to balance societal, individual, and industry interests while minimizing legal and ethical risks [74].

The keywords “privacy” and “technology” both appear 12 times. These keywords indicate growing attention to the challenges of balancing the protection of individual privacy with the need for collaborative data practices in the realm of intellectual property. For example, Ahire, P. studied a secure cloud model to protect the intellectual privacy of arithmetic expressions in source codes by employing data obfuscation techniques [75].

Legal foundations are represented by the keyword “law”, which emphasizes the importance of legal frameworks in shaping the discourse on intellectual property rights. Finally, “data mining”, although a relatively recent and less frequent keyword, highlights the growing relevance of advanced analytical techniques in the management and protection of data-based IPR, illustrating the field’s adaptation to the complexities of the digital age. Manteghi, M. argues that the current EU regulations regarding text and data mining (TDM) fail to balance the fundamental right to information and research with intellectual property rights, proposing legal amendments to improve this balance to allow for innovation across the EU [76].

3.4.2. Research Hotspots Analysis

CiteSpace’s Time Zone View (Figure 9) visually delineates the temporal evolution of scholarly publications or patents, revealing research trends and thematic shifts over time.

Initially, the keyword “intellectual property rights” emerged as a central node in 1997, indicating the inception of focused discussions on the protection and management of rights within the digital sphere. Alongside this, “intellectual property protection” became a dominant theme, suggesting an early emphasis on establishing legal frameworks to safeguard intangible assets in the data-driven economy [62].

As the field grew, the term ‘innovation’ became more prominent after 2005. This period saw an increase in patent filings and a greater awareness of how strong IP protection mechanisms can incentivize innovation.

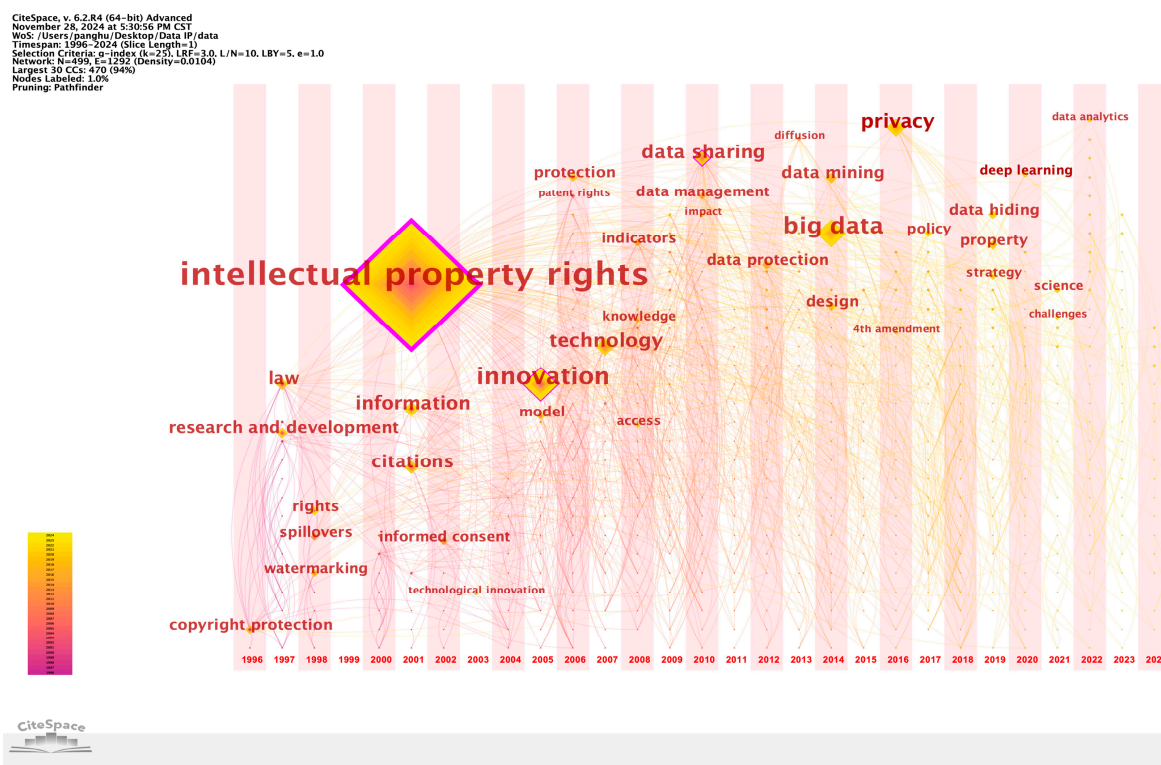


Figure 9. Keyword time zones.

With the advent of the big data, the incidence of the keywords ‘big data’ and ‘data sharing’ increased in 2010. The emergence of ‘data protection’ in 2012 and ‘privacy’ highlighted a growing awareness of the need to protect personal information, reflecting a growing emphasis on data security and individual rights. Several scholars have discussed the 10-year data protection period, emphasizing its importance in protecting clinical and safety data for pharmaceutical and chemical products, while arguing for legal reforms to better balance incentives for innovation and regulatory fairness [77,78]. Liaw, S. T. highlighted the challenges in ensuring data privacy in the management of Electronic Health Records (EHRs), proposing solutions such as transparent governance, ethical access, and privacy protection to ensure the secure use of health data [79].

Technological advances are also indicated by the emergence of the terms ‘data mining’ and ‘data analytics’. The more recent emergence of the term ‘data hiding’ suggests an active exploration of data security techniques in response to the increasing threats from the internet and the need for more complicated methods of data hiding.

In summary, it is clear that research on data-based intellectual property rights has broadened and deepened over time. The basic concepts remain. However, the discourse broadened to include new challenges posed by technological innovations and the growing digital economy.

3.4.3. Keyword Burst Analysis

Keyword bursts help us to identify keywords that suddenly increase in frequency within a given time period, which may represent an emerging research trend or hotspot. Figure 10 shows the top 10 keywords that experienced the most significant surges in prominence from 1996 to 2024. The red line indicates the period during which the keyword was in a state of burst, while the dark blue line indicates the period of the keyword’s emergence. A higher “strength” value indicates a greater likelihood that the keyword will become a leading topic in the field. The terms “begin” and “end” represent the start and end of each keyword’s burst phase, respectively.

As shown in Figure 10, the Citespace analysis pinpointed key periods of intense scholarly focus on the management and protection of data-based IP. Early attention focused on “patent rights” and “data protection”, laying the legal groundwork for the use of data as a form of intellectual property. Subsequently, there was a surge in researcher interest in ‘technology’ and evaluative ‘metrics’, reflecting a shift in methodology towards technological solutions and quantitative assessments. The term “privacy” experienced the longest significant burst, indicating a continued concern about the increased awareness of data privacy and data breaches. The burst of “data mining” and “big data” buzzwords indicates that how to fully exploit the value of big data while protecting privacy and security is an important topic in current scientific research and the development of technology.

Top 10 Keywords with the Strongest Citation Bursts

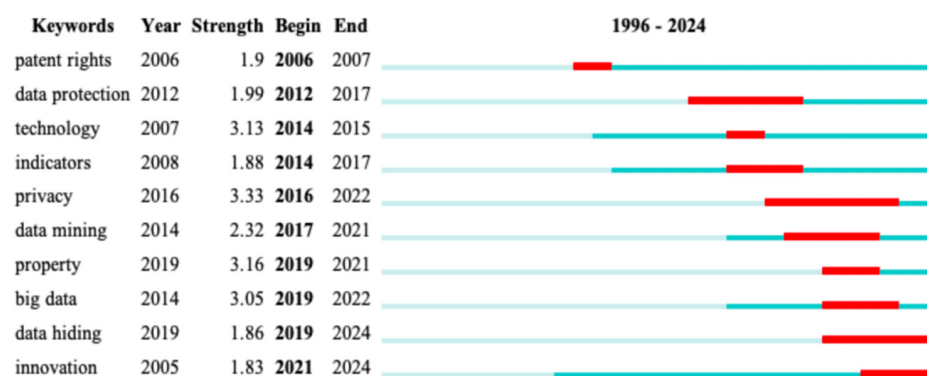


Figure 10. Top 10 keywords with strong bursts.

The most recent burst keywords are ‘property’, ‘data hiding’, and ‘innovation’, suggesting a current preoccupation with defining data property rights and deploying innovative data protection techniques. The continued attention to ‘innovation’ and the emergence of ‘data hiding’ highlight a desire to promote creativity and protect intellectual property in the digital age.

3.5. Knowledge Framework

As shown in Figure 11, publication analysis helps demonstrate the popularity of research in the field and its potential growth, guiding researchers to invest further in its exploration. The analysis of the top three journals in the field indicates a diversified interest in the technological, legal, and ethical dimensions of the management and protection of data-based intellectual property.

1. *Lecture Notes in Computer Science*, *Multimedia Tools and Applications*, and *Computer Law Security Review* are the three most prominent journals regarding this area of research. The categories of “Computer Science”, “Information Systems”, and “Law” receive the most focus.
2. Collaboration analysis represents the dynamic cooperation relationships among institutions, regions, and authors; we list the leading cooperative participants in the field, aiming to provide researchers with a reference when looking for potential collaborative partners and allowing for them to determine appropriate collaborative research areas. Specifically, leading cooperative institutions include the University of California System, the University of London, and the National Institute of Technology (NIT System). The USA, the People’s Republic of China, and England are the most active regions of collaboration. Guo Xiaolong, Li Teng, and Dutta Raj Gautam emerged as the most active collaborative authors.
3. Co-citation indicates the co-citation status of authors, journals, and references; this highlights the most referential and significant contributions in the field, aiming to provide researchers with a clearer understanding of the current evolution in the

management and protection of data-based intellectual property, as well as important areas for future research. The top three most co-cited authors are Jaffe AB, Hall BH, and Samuelson P. Prominent co-cited journals include *RES Policy*, *AM ECON REV*, and *Science*. The top three most co-cited references are Wolfert S (2017), Wilkinson MD (2016), and Mowery DC (2004) [53,54,59]. Different collaborators have different research areas; scholars can look to their own research interests and find opportunities for future collaboration.

4. The keyword co-occurrence analysis highlights the critical issues and trends in the field and suggests possible future research directions. Core themes include intellectual property rights, innovation, and big data, demonstrating the central focus on the management and protection of data in a rapidly developing digital society. Recent hotspots involve topics such as data mining, data analytics, and privacy, indicating the growing emphasis on extracting valuable insights from data while maintaining security and privacy protections. Additionally, burst keywords like big data, data hiding, and innovation suggest emerging trends in leveraging technological advancements to address complex challenges pertaining to data-based intellectual property. These developments provide important insights for researchers looking to explore the intersection of technology, data security, and intellectual property rights in the future.

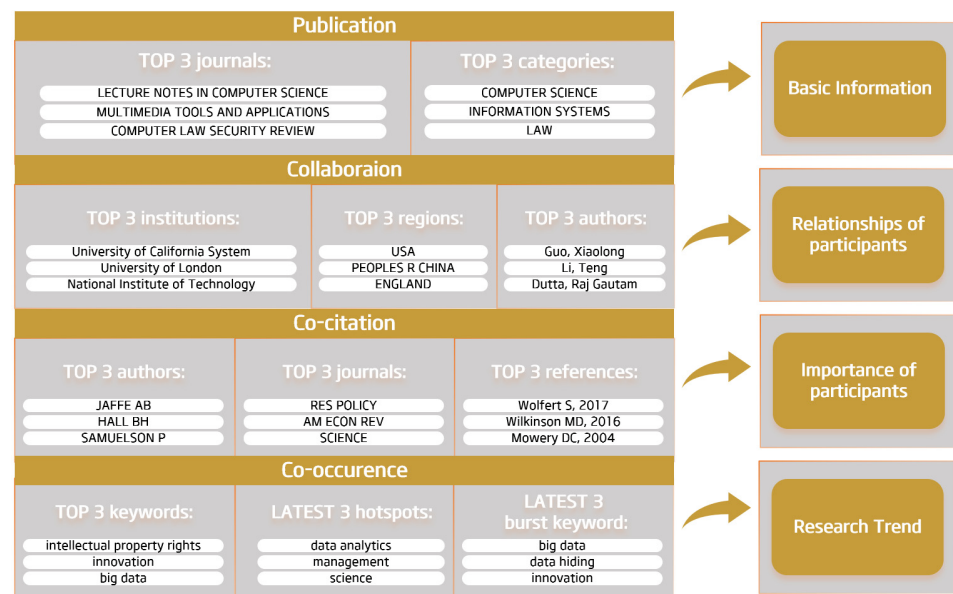


Figure 11. Knowledge framework [53,54,59].

4. Discussion

4.1. Current Challenges

In the above analysis, the management and protection of data-based intellectual property (IP) are shown to face several distinct challenges, particularly due to rapid technological advancements and varying global regulations. Scholars and practitioners in this field should focus on the following key areas.

1. Global inconsistencies in data protection regulations: The landscape of data-based intellectual property (IP) is heavily influenced by the disparities in data protection regulations across regions. While frameworks like the General Data Protection Regulation (GDPR) provide comprehensive protection in the European Union, many other regions lack equivalent standards. This leads to fragmentation in legal environments and creates uncertainty for businesses managing data globally [80].
2. Cybersecurity threats and data breaches: With the increasing number of data breaches and cyberattacks, protecting proprietary data has become a critical challenge [42]. Businesses face the dual challenge of ensuring data security while adhering to complex IP laws.

3. Emerging technologies and complex ownership models: Technologies such as blockchain and artificial intelligence (AI) have introduced new complexities in the definition and protection of data ownership. These technologies often operate in decentralized networks to which traditional intellectual property frameworks may struggle to adapt. Managing IP in these environments requires updated legal and operational approaches [25].
4. Balancing data ownership and privacy: As data are increasingly sourced from multiple entities, determining ownership rights has become more complicated. There is a need to balance the protection of personal privacy with the promotion of data-sharing to allow for innovation and scientific research. Finding a way to protect privacy while facilitating data use is an ongoing challenge [81].

4.2. Identified Future Research

As data-based intellectual property research continues to evolve, future studies will likely focus on new and emerging trends that address the current challenges. Based on the existing knowledge framework and the challenges identified above, we propose the following future research directions.

1. There will be an increasing focus on harmonizing global data protection standards to create a cohesive legal framework. Scholars may discuss how personal genomic data, including intellectual property rights and ownership, intersect with international agreements such as the TRIPS Agreement, affecting trade and legal practices in data use and protection. This focus on standardization is expected to dominate future research programs as more organizations demand a consistent global approach [31].
2. Blockchain and AI are likely to remain central to future data-based IP research. Studies will explore how distributed systems such as blockchain can be used to enforce IP rights, and how smart contracts can automate the process of verifying data ownership. Wang, F. F. has already laid the groundwork for understanding the intersection between emerging technologies and IP protection [25].
3. Future research should focus on creating more adaptable and flexible frameworks for intellectual property protection, as technology is evolving faster than existing legal regimes. To ensure that data ownership and intellectual property rights are protected in a changing technological landscape, Leistner argues that legal structures should evolve dynamically to keep pace with changes in artificial intelligence and big data analytics [6].
4. The complexity of data ownership by individuals, businesses, and governments requires a multi-stakeholder governance framework. Future research may explore how to integrate these different stakeholders into a coherent governance structure. Saulnier discussed the benefits and challenges of designing coordinated access protocols for international data-sharing consortia. They suggest that, particularly in the context of large-scale genomic data-sharing, a multi-stakeholder governance model could address intellectual property rights issues [82]. Timotijevic argues that balancing the interests of multiple stakeholders is key to establishing a clear data ownership model [83]. Research in this area will focus on collaborative models that ensure the fair and equitable management of data-based intellectual property rights.

5. Conclusions

This bibliometric analysis of the management and protection of data-based intellectual property provides several key insights. (1) Our analysis reveals an upward trend in the number of publications, indicating a growing scholarly interest in data-based IP as digital technologies evolve. (2) Through a collaboration analysis, we found that institutions from the United States, China, and the United Kingdom are leading the global research efforts, although collaboration between institutions remains fragmented. (3) A co-citation analysis uncovered the foundational literature in this domain, with works by Jaffe AB, Hall BH, and Samuelson P playing critical roles in shaping the discourse. (4) A co-occurrence analysis of keywords such as “innovation”, “big data”, and “privacy” highlighted emerg-

ing themes, with growing attention being paid to data privacy, technological solutions, and regulatory frameworks.

This study makes several notable contributions. Firstly, by analyzing collaborative networks and co-citations, this study provides a visual study of data-based intellectual property research. Secondly, it builds a comprehensive knowledge framework that guides scholars to the key literature and emerging research areas. Finally, this study provides information to researchers and policy makers by identifying current research hotspots and potential future trends, such as the integration of data-based IP with AI and blockchain technologies.

However, the study is not without its limitations. (1) Our focus was exclusively on English-language publications, to the potential exclusion of critical contributions in other languages. (2) The dataset was limited to articles indexed in WoS, which may have omitted important studies from other databases. (3) The analysis was limited to publications from up to 2024; ongoing developments in the field may have an impact on future research directions. Future studies should expand this dataset and explore non-English sources to provide a more holistic understanding of the global data IP landscape.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/info15120780/s1>.

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